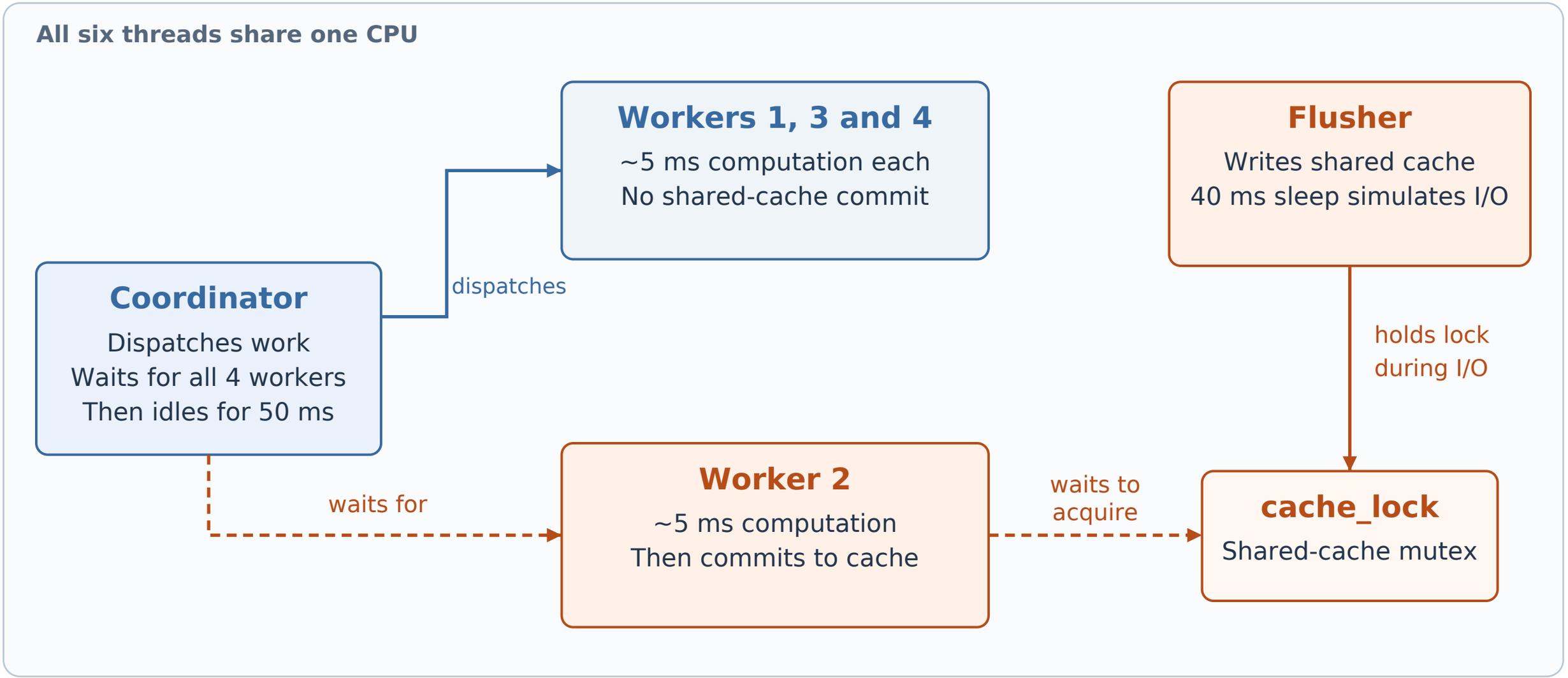

All six threads share one CPU
Workers 1, 3 and 4
~5 ms computation each
No shared-cache commit
Flusher
Writes shared cache
40 ms sleep simulates I/O
Coordinator
Dispatches work
Waits for all 4 workers
Then idles for 50 ms
dispatches
holds lock
during I/O
waits for
Worker 2
~5 ms computation
Then commits to cache
waits to
acquire
cache_lock
Shared-cache mutex
Waiting dependency
Lock contention delays worker 2 and the entire round.